
Impact on Small Business Loans During the 2008 Recession

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Abstract

We aim to look at small business loans sent out between 2007-2012, and see if we can determine if ZIP code increases the likelihood of loans being approved and repaid. This is an important problem for us as this was the last major economic recession in US history, and as graduate students we are curious to know what may be a good way to ride out another potential recession.

1 Introduction

Many people have a fear of a recession as of late, with J.P. Morgan Chase estimating a 40 percent chance of the U.S. economy entering a recession by the end of 2025 (<https://money.usnews.com/investing/articles/recession-2025-what-to-watch-how-to-prepare>). As graduate students, we thought it would be a productive experience to leverage our machine learning knowledge and apply it to a dataset of small business loans, specifically looking at the period of 2007-2012 during the 2008 Recession. This will hopefully give us an edge on applying for jobs, by potentially identifying zip codes that were resilient to the effects of a recession, better helping with our job searches in the future.

To do: add high level explanation of ML approach that we choose to use.

2 Related Works

Deep Learning for mortgage risk explores client behavior patterns by modeling payment dynamics across regions and at the ZIP code level. The study primarily focused on tracking prepayment, foreclosure, and delinquency states.

Using loan-level data from 2005-2011, *Small Business Lending and the Great Recession* analyses the lending patterns to small businesses across regions during the financial crisis.

3 Methodology

We will be preprocessing the data set located at <https://www.kaggle.com/datasets/mirbektoktogaraev/should-this-loan-be-approved-or-denied/data>, and we will be running experiments on an ipynb file. We cleaned the data and performed feature engineering. We then split the data into training and validation sets, and scaled and ran it for different ML models. We will then compare and contrast the different data sets and choose one to run on the test set.

4 Experiments

We ran a variety of different models to try to find the best performing model. With one model, we ran One Hot Encoding to encode whether or not a loan was repaid, and utilized HistGradientBoostingClassifier. This model also cut the dataset to focus specifically between 2007 and 2010, during

the height of the recession. We ultimately decided not to use this model for our final analysis, as although it was accurate, we thought it overly sampled the failures.

The second model also utilized OHE, but instead of a gradient descent, used a logistical regression model. This achieved fair accuracy of about 85 percent, but wasn't what we ended up utilizing. It did show though, that a simple model could effectively work with and fit our data.

Lastly, the model we chose to work with, used an Adaboosted decision tree with tuned hyperparameters, experimenting with different depths and the number of minimum samples per leaf. This model had high accuracy, comparable to the HistGradientBoostingClassifier model, while still sampling the full date range we wanted to look at. We found best accuracy at a depth of 9, and a min leaf sample of 100.

5 Results

The logistic regression model resulted in a ROC-AUC of 0.859 for the validation set. We had much higher rates of ROC-AUC for the other models, and noticed that all models struggled at predicting charged off loans, reflected in their lower precision and recall scores. Since all models were able to identify paid in full loans with relative ease, we chose the precision and recall of charge off loans as our target metrics.

The decision trees showed increased precision and recall scores for higher depths, but the gains began to level off around a depth of 9. With a max depth of 9, we saw a chg off precision of 83 and a recall of 78.

Adaboost provided precision and recalls scores of 86 and 83 with less variance than the high depth decision trees. We choose the Adaboost as the model with the best variance/bias tradeoff. It's performance on the test set was nearly identical to the validation set, suggesting that this model would perform well on other data sets, as the test set was only used at the end, and the hyper parameter and model selection did not involve the test set.

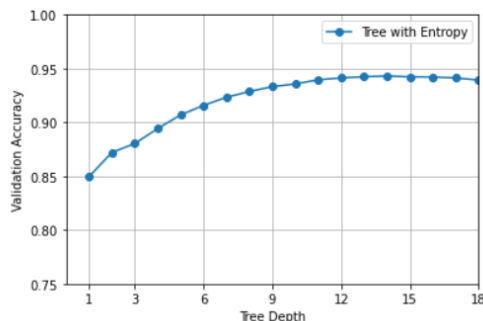


Figure 1: Comparison of tree depths to prevent over or underfitting.

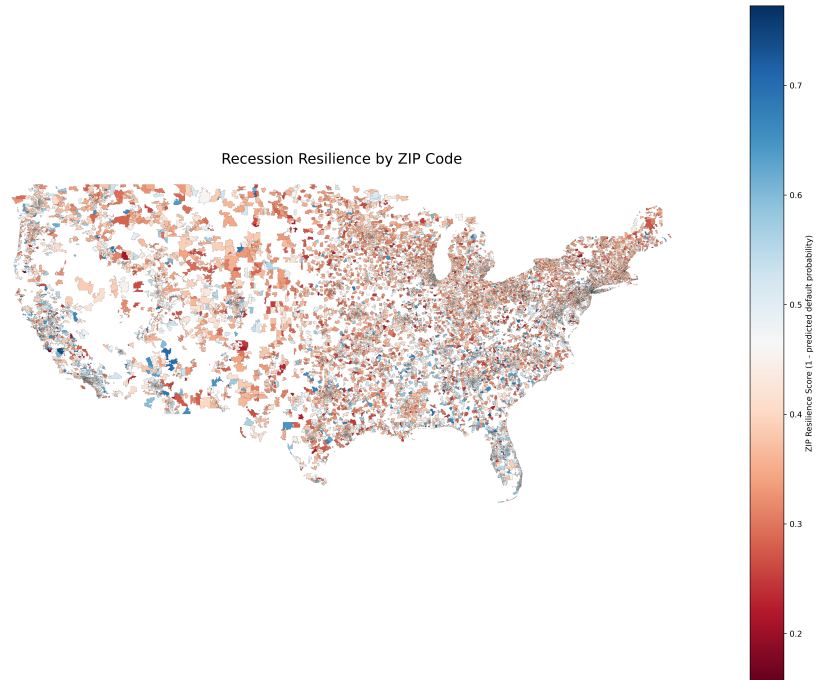


Figure 2: A heatmap of zipcodes, colored by resilience.

Above we created a figure using the recession resilience scores for each zip code in the continental United States. Resilience score was computed by taking the average predicted default rate for a zip code, and subtracting it from 1. Zip codes that had less than 20 loans during the recession period (2007-2012) were excluded from analysis, and represent the blank areas on the map. This leads us to believe areas that are colored blue are more resilient against a recession, while areas that are colored red are less resilient against a recession. This visualization suggests that rural areas are less resilient versus recessions, while urban areas are more resilient.

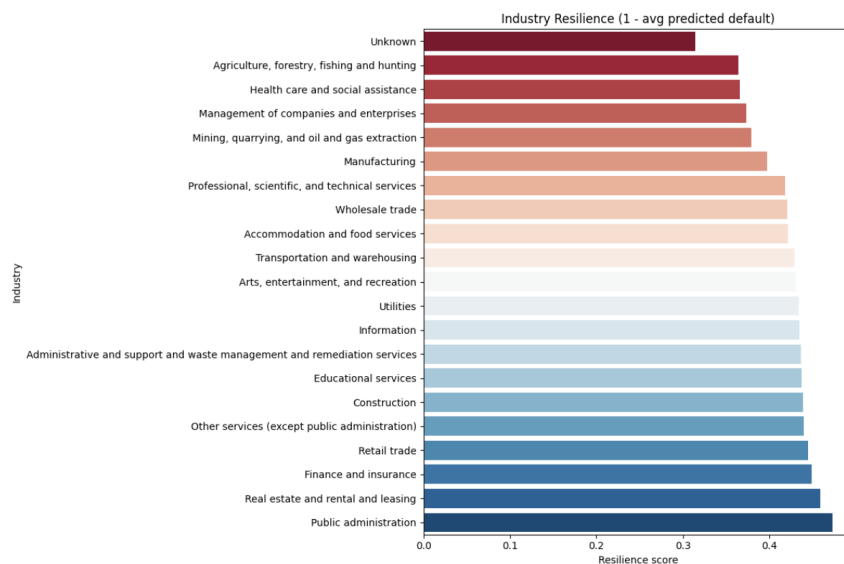


Figure 3: Industries ranked by resilience score.

After performing our zip code analysis, we create the figure above which breaks down predicted defaults based on industry type. While looking at the results, we were concerned about the possible conflation between industries and their zip code. For example, farms only exist in predominantly

rural zip codes, while not existing in urban zip codes. These results tend to support this concern, as more rural associated industries are found to be overwhelmingly less resilient against a recession (Agriculture, forestry, fishing and hunting). Meanwhile, the strongest industries against recession is public administration, real estate, and finance. However there are many possible reasons as to why this may be the case, ranging from government support of these industries to they inherently being strong against recessions - such is outside the scope of this study.

6 Conclusion

This study sought to analyze small business lending outcomes during the 2008 recession by developing machine learning models that were capable of estimating the risk of default, and how that risk was potentially exacerbated by their geographic region. We evaluated several different models for use, including logistic regression, gradient boosted decision trees, and AdaBoost. This suggests that tree-based interpretations can capture complex relationships between borrowers, loan attributes, and the macroeconomic pressures present during the recession period.

Our zip code analysis shows a clear rural / urban divide recession resilience, supported by our analysis of industries which shows that industries that are more associated with rural areas are more likely to default on their loans compared to industries associated with urban areas. Although our model highlights this correlation, further work would need to be done to more closely analyze how these differences came about.

Altogether, this underscores the importance of geography and industry when looking to understand the performance of small business loans during periods of economic distress. As concerns about future economic downturns persist, analyses of this type may offer valuable guidance for policymakers, lenders, and individuals seeking to better understand the distribution of economic risk.

7 Acknowledgements

Cole, Rebel A., and Tatyana Sokolyk. "How Do Banks Respond to Increased Funding Uncertainty? Evidence from the U.S. Mortgage and Small Business Lending Crisis." 2012.

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